Bellabeat Case Study - Google Data Analytics Certificate

Charles Alders

Ask

- Business Task
  - Analyze smart device usage for non-Bellabeat smart devices and apply insights to Bellabeat devices.
- Stakeholders
  - Urška Sršen: Bellabeat's cofounder and Chief Creative Officer
  - Sando Mur: Mathematician and Bellabeat's cofounder; key member of the Bellabeat executive team
  - Bellabeat marketing analytics team: A team of data analysts responsible for collecting, analyzing, and reporting data that helps guide Bellabeat's marketing strategy
  - The Customer: after all, the customer is the one who will be receiving the benefits from the data insight.

Prepare

- The data:
  - is hosted on kaggle.com & other data-sharing platforms.
  - contains 18 .csv files that are all presented in long format.
  - is licensed under CC0: Public Domain, so it is open-source.
  - from a credible source, https://zenodo.org/record/53894#.YMoUpnVKiP9 (https://zenodo.org/record/53894#.YMoUpnVKiP9)
- It is worth noting that this data spans just 2 months in 2016 and includes Fitbit data from 30 users, so a very small sample size.

Process

```
library(tidyverse)
```

```
## ── Attaching packages ─────────────────────────────────────── tidyverse 1.3.2 ──
## ✔ ggplot2 3.4.1     ✔ purrr   1.0.1
## ✔ tibble  3.1.8     ✔ dplyr   1.1.0
## ✔ tidyr   1.3.0     ✔ stringr 1.5.0
## ✔ readr   2.1.4     ✔ forcats 1.0.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
```

```
library(here)
```

```
## here() starts at /Users/charliealders/Desktop/Capstone
```

```
library(skimr)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'
##
## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

I am going to start by loading the activity and sleep tables, as those seem to be the most relevant to the analysis for now. Below shows a preview of both data frames.

```
activity <- read_csv("FitabaseData/dailyActivity_merged.csv", show_col_types = FALSE)
sleep <- read_csv("FitabaseData/sleepDay_merged.csv", show_col_types = FALSE)
head(activity)
```

```
## # A tibble: 6 × 15
##       Id Activ…¹ Total…² Total…³ Track…⁴ Logge…⁵ VeryA…⁶ Moder…⁷ Light…⁸ Seden…⁹
##    <dbl> <chr>     <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 1.50e9 4/12/2…   13162    8.5     8.5         0    1.88   0.550    6.06       0
## 2 1.50e9 4/13/2…   10735    6.97    6.97        0    1.57   0.690    4.71       0
## 3 1.50e9 4/14/2…   10460    6.74    6.74        0    2.44   0.400    3.91       0
## 4 1.50e9 4/15/2…    9762    6.28    6.28        0    2.14   1.26     2.83       0
## 5 1.50e9 4/16/2…   12669    8.16    8.16        0    2.71   0.410    5.04       0
## 6 1.50e9 4/17/2…    9705    6.48    6.48        0    3.19   0.780    2.51       0
## # … with 5 more variables: VeryActiveMinutes <dbl>, FairlyActiveMinutes <dbl>,
## #   LightlyActiveMinutes <dbl>, SedentaryMinutes <dbl>, Calories <dbl>, and
## #   abbreviated variable names ¹ActivityDate, ²TotalSteps, ³TotalDistance,
## #   ⁴TrackerDistance, ⁵LoggedActivitiesDistance, ⁶VeryActiveDistance,
## #   ⁷ModeratelyActiveDistance, ⁸LightActiveDistance, ⁹SedentaryActiveDistance
```

```
head(sleep)
```

```
## # A tibble: 6 × 5
##           Id SleepDay              TotalSleepRecords TotalMinutesAsleep TotalT…¹
##        <dbl> <chr>                             <dbl>              <dbl>    <dbl>
## 1 1503960366 4/12/2016 12:00:00 AM                 1                327      346
## 2 1503960366 4/13/2016 12:00:00 AM                 2                384      407
## 3 1503960366 4/15/2016 12:00:00 AM                 1                412      442
## 4 1503960366 4/16/2016 12:00:00 AM                 2                340      367
## 5 1503960366 4/17/2016 12:00:00 AM                 1                700      712
## 6 1503960366 4/19/2016 12:00:00 AM                 1                304      320
## # … with abbreviated variable name ¹TotalTimeInBed
```

A quick preview to see a summary of the data frame.

```
skim_without_charts(activity)
```

Data summary

Name	activity
Number of rows	940
Number of columns	15
_____	
Column type frequency:	
character	1
numeric	14
_____	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
ActivityDate	0	1	8	9	0	31	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	
Id	0	1	4.855407e+09	2.424805e+09	1503960366	2.320127e+09	4.445115e+09	6.962181e+09	8.877
TotalSteps	0	1	7.637910e+03	5.087150e+03	0	3.789750e+03	7.405500e+03	1.072700e+04	3.601
TotalDistance	0	1	5.490000e+00	3.920000e+00	0	2.620000e+00	5.240000e+00	7.710000e+00	2.803
TrackerDistance	0	1	5.480000e+00	3.910000e+00	0	2.620000e+00	5.240000e+00	7.710000e+00	2.803
LoggedActivitiesDistance	0	1	1.100000e-01	6.200000e-01	0	0.000000e+00	0.000000e+00	0.000000e+00	4.940

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	
VeryActiveDistance	0	1	1.500000e+00	2.660000e+00	0	0.000000e+00	2.100000e-01	2.050000e+00	2.192
ModeratelyActiveDistance	0	1	5.700000e-01	8.800000e-01	0	0.000000e+00	2.400000e-01	8.000000e-01	6.480
LightActiveDistance	0	1	3.340000e+00	2.040000e+00	0	1.950000e+00	3.360000e+00	4.780000e+00	1.071
SedentaryActiveDistance	0	1	0.000000e+00	1.000000e-02	0	0.000000e+00	0.000000e+00	0.000000e+00	1.100
VeryActiveMinutes	0	1	2.116000e+01	3.284000e+01	0	0.000000e+00	4.000000e+00	3.200000e+01	2.100
FairlyActiveMinutes	0	1	1.356000e+01	1.999000e+01	0	0.000000e+00	6.000000e+00	1.900000e+01	1.430
LightlyActiveMinutes	0	1	1.928100e+02	1.091700e+02	0	1.270000e+02	1.990000e+02	2.640000e+02	5.180
SedentaryMinutes	0	1	9.912100e+02	3.012700e+02	0	7.297500e+02	1.057500e+03	1.229500e+03	1.440
Calories	0	1	2.303610e+03	7.181700e+02	0	1.828500e+03	2.134000e+03	2.793250e+03	4.900

```
skim_without_charts(sleep)
```

Data summary

Name	sleep
Number of rows	413
Number of columns	5
_____	
Column type frequency:	
character	1
numeric	4
_____	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
SleepDay	0	1	20	21	0	31	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Id	0	1	5.000979e+09	2.06036e+09	1503960366	3977333714	4702921684	6962181067	8792009665
TotalSleepRecords	0	1	1.120000e+00	3.50000e-01	1	1	1	1	3
TotalMinutesAsleep	0	1	4.194700e+02	1.18340e+02	58	361	433	490	796
TotalTimeInBed	0	1	4.586400e+02	1.27100e+02	61	403	463	526	961

Here we can see that there are no missing values. As we saw from the head() of the dataframes, there are many 0's. These are unavoidable, as these people likely did not perform certain types of exercise. There are too many to drop the rows, and it would skew the data if I replaced them with means/medians, so I will work around them by filtering them out in calculations.

```
activity <- clean_names(activity)
sleep <- clean_names(sleep)
```

Cleaned the column names for both dataframes to be standardized.

```
n_unique(activity$id)
```

```
## [1] 33
```

```
n_unique(sleep$id)
```

```
## [1] 24
```

The sample size for the data is quite low. It is not recommended to proceed with a sample size <30, but I am goint to continue for practice!

```
sum(duplicated(activity))
```

```
## [1] 0
```

```
sum(duplicated(sleep))
```

```
## [1] 3
```

There are 3 duplicated rows in the sleep table. Let's drop them.

```
sleep <- sleep %>% distinct()
sum(duplicated(sleep))
```

```
## [1] 0
```

No more duplicates. This would be a good time to remove NULL values as well, but our dataframes have 0s for missing.

```
activity[2]
```

```
## # A tibble: 940 × 1
##    activity_date
##    <chr>
##  1 4/12/2016
##  2 4/13/2016
##  3 4/14/2016
##  4 4/15/2016
##  5 4/16/2016
##  6 4/17/2016
##  7 4/18/2016
##  8 4/19/2016
##  9 4/20/2016
## 10 4/21/2016
## # … with 930 more rows
```

```
sleep[2]
```

```
## # A tibble: 410 × 1
##    sleep_day
##    <chr>
##  1 4/12/2016 12:00:00 AM
##  2 4/13/2016 12:00:00 AM
##  3 4/15/2016 12:00:00 AM
##  4 4/16/2016 12:00:00 AM
##  5 4/17/2016 12:00:00 AM
##  6 4/19/2016 12:00:00 AM
##  7 4/20/2016 12:00:00 AM
##  8 4/21/2016 12:00:00 AM
##  9 4/23/2016 12:00:00 AM
## 10 4/24/2016 12:00:00 AM
## # … with 400 more rows
```

The date formats for both tables are different, so it would be best to make them consistent. Dropping the time from sleep makes the most sense to me, as the times are all set to 12AM.

```
activity <- activity %>%
  rename(date = activity_date) %>%
  mutate(date = as_date(date, format = "%m/%d/%Y"))

sleep <- sleep %>%
  rename(date = sleep_day) %>%
  mutate(date = as_date(date, format = "%m/%d/%Y %I:%M:%S %p"))

activity[2]
```

```
## # A tibble: 940 × 1
##    date
##    <date>
##  1 2016-04-12
##  2 2016-04-13
##  3 2016-04-14
##  4 2016-04-15
##  5 2016-04-16
##  6 2016-04-17
##  7 2016-04-18
##  8 2016-04-19
##  9 2016-04-20
## 10 2016-04-21
## # … with 930 more rows
```

```
sleep[2]
```

```
## # A tibble: 410 × 1
##    date
##    <date>
##  1 2016-04-12
##  2 2016-04-13
##  3 2016-04-15
##  4 2016-04-16
##  5 2016-04-17
##  6 2016-04-19
##  7 2016-04-20
##  8 2016-04-21
##  9 2016-04-23
## 10 2016-04-24
## # … with 400 more rows
```

Now that the date columns are normalized, I'll merge the two data frames so it is easy to compare activity and sleep.

```
df <- merge(activity, sleep, by=c("id", "date"), all.x = TRUE, all.y = TRUE)
head(df)
```

```
##           id       date total_steps total_distance tracker_distance
## 1 1503960366 2016-04-12       13162           8.50             8.50
## 2 1503960366 2016-04-13       10735           6.97             6.97
## 3 1503960366 2016-04-14       10460           6.74             6.74
## 4 1503960366 2016-04-15        9762           6.28             6.28
## 5 1503960366 2016-04-16       12669           8.16             8.16
## 6 1503960366 2016-04-17        9705           6.48             6.48
##   logged_activities_distance very_active_distance moderately_active_distance
## 1                          0                 1.88                       0.55
## 2                          0                 1.57                       0.69
## 3                          0                 2.44                       0.40
## 4                          0                 2.14                       1.26
## 5                          0                 2.71                       0.41
## 6                          0                 3.19                       0.78
##   light_active_distance sedentary_active_distance very_active_minutes
## 1                  6.06                         0                  25
## 2                  4.71                         0                  21
## 3                  3.91                         0                  30
## 4                  2.83                         0                  29
## 5                  5.04                         0                  36
## 6                  2.51                         0                  38
##   fairly_active_minutes lightly_active_minutes sedentary_minutes calories
## 1                    13                    328               728     1985
## 2                    19                    217               776     1797
## 3                    11                    181              1218     1776
## 4                    34                    209               726     1745
## 5                    10                    221               773     1863
## 6                    20                    164               539     1728
##   total_sleep_records total_minutes_asleep total_time_in_bed
## 1                   1                  327               346
## 2                   2                  384               407
## 3                  NA                   NA                NA
## 4                   1                  412               442
## 5                   2                  340               367
## 6                   1                  700               712
```

```
nrow(df)
```

```
## [1] 940
```

With the new dataframe merge, there are some missing values. I'll replace the NAs with 0s since the other missing values are already 0s.

```
df[is.na(df)] <- 0
sum(is.na(df))
```

```
## [1] 0
```

Looking at the data, it looks like there are a lot of missing (0) values in the logged_activities_distance and the sedentary_active_distance columns.

```
df %>%
  filter(logged_activities_distance > 0) %>%
  count(sum(logged_activities_distance))
```

```
##   sum(logged_activities_distance)  n
## 1                        101.6807 32
```

```
df %>%
  filter(sedentary_active_distance > 0) %>%
  count(sum(sedentary_active_distance))
```

```
##   sum(sedentary_active_distance)  n
## 1                           1.51 82
```

Only 32 and 82 entries (out of 940) respectfully. This is just taking up extra space, so lets remove it.

```
df <- subset(df, select = -c(logged_activities_distance, sedentary_active_distance))
head(df)
```

```
##           id       date total_steps total_distance tracker_distance
## 1 1503960366 2016-04-12       13162           8.50             8.50
## 2 1503960366 2016-04-13       10735           6.97             6.97
## 3 1503960366 2016-04-14       10460           6.74             6.74
## 4 1503960366 2016-04-15        9762           6.28             6.28
## 5 1503960366 2016-04-16       12669           8.16             8.16
## 6 1503960366 2016-04-17        9705           6.48             6.48
##   very_active_distance moderately_active_distance light_active_distance
## 1                 1.88                       0.55                  6.06
## 2                 1.57                       0.69                  4.71
## 3                 2.44                       0.40                  3.91
## 4                 2.14                       1.26                  2.83
## 5                 2.71                       0.41                  5.04
## 6                 3.19                       0.78                  2.51
##   very_active_minutes fairly_active_minutes lightly_active_minutes
## 1                  25                    13                    328
## 2                  21                    19                    217
## 3                  30                    11                    181
## 4                  29                    34                    209
## 5                  36                    10                    221
## 6                  38                    20                    164
##   sedentary_minutes calories total_sleep_records total_minutes_asleep
## 1               728     1985                   1                  327
## 2               776     1797                   2                  384
## 3              1218     1776                   0                    0
## 4               726     1745                   1                  412
## 5               773     1863                   2                  340
## 6               539     1728                   1                  700
##   total_time_in_bed
## 1               346
## 2               407
## 3                 0
## 4               442
## 5               367
## 6               712
```

```
df <- df %>%
  rename(times_asleep = total_sleep_records, total_mins_in_bed = total_time_in_bed)
```

Now that the dataframe is cleaned and merged, it's time for analysis!

Analyze

Below I created a new column, which represents the percent of time spent inactive (out of the tracked time on the Fitbit device). Next, I reordered the column in a logical order.

```
df <- df %>%
  mutate(percent_sedentary = sedentary_minutes/(very_active_minutes + fairly_active_minutes + lightly_active_minu
tes + sedentary_minutes)*100)

df <- df[, c(1:12, 17, 13:16)]
```

Additionally, I wanted to see how much extra time people spend in bed while they are awake, so I added another column to represent the difference between time asleep and in bed.

```
df <- df %>%
  mutate(extra_mins_in_bed = total_mins_in_bed - total_minutes_asleep)
head(df)
```

```
##           id       date total_steps total_distance tracker_distance
## 1 1503960366 2016-04-12       13162           8.50             8.50
## 2 1503960366 2016-04-13       10735           6.97             6.97
## 3 1503960366 2016-04-14       10460           6.74             6.74
## 4 1503960366 2016-04-15        9762           6.28             6.28
## 5 1503960366 2016-04-16       12669           8.16             8.16
## 6 1503960366 2016-04-17        9705           6.48             6.48
##   very_active_distance moderately_active_distance light_active_distance
## 1                 1.88                       0.55                  6.06
## 2                 1.57                       0.69                  4.71
## 3                 2.44                       0.40                  3.91
## 4                 2.14                       1.26                  2.83
## 5                 2.71                       0.41                  5.04
## 6                 3.19                       0.78                  2.51
##   very_active_minutes fairly_active_minutes lightly_active_minutes
## 1                  25                    13                    328
## 2                  21                    19                    217
## 3                  30                    11                    181
## 4                  29                    34                    209
## 5                  36                    10                    221
## 6                  38                    20                    164
##   sedentary_minutes percent_sedentary calories times_asleep
## 1               728          66.54479     1985            1
## 2               776          75.12101     1797            2
## 3              1218          84.58333     1776            0
## 4               726          72.74549     1745            1
## 5               773          74.32692     1863            2
## 6               539          70.82786     1728            1
##   total_minutes_asleep total_mins_in_bed extra_mins_in_bed
## 1                  327               346                19
## 2                  384               407                23
## 3                    0                 0                 0
## 4                  412               442                30
## 5                  340               367                27
## 6                  700               712                12
```

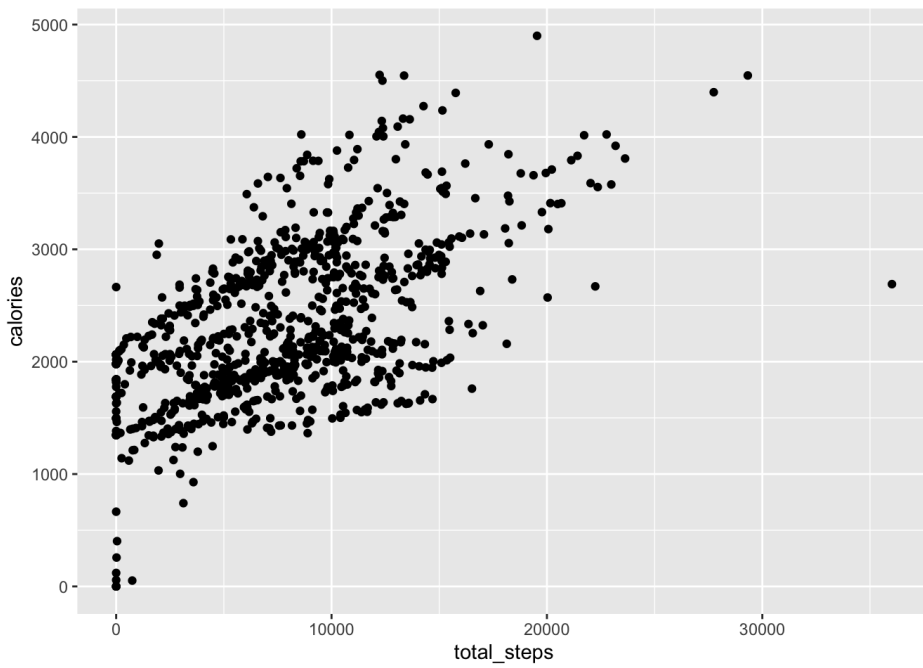
```
df %>%
  select(total_steps,
         total_distance,
         sedentary_minutes,
         total_minutes_asleep,
         total_mins_in_bed) %>%
  filter(total_steps!=0, total_distance!=0, sedentary_minutes!=0, total_minutes_asleep!=0, total_mins_in_bed!=0)
%>%
  summary()
```

```
##   total_steps    total_distance   sedentary_minutes total_minutes_asleep
##  Min.   :   42   Min.   : 0.030   Min.   :   2.0    Min.   : 58.0
##  1st Qu.: 5206   1st Qu.: 3.600   1st Qu.: 632.0    1st Qu.:361.0
##  Median : 8915   Median : 6.280   Median : 717.0    Median :433.0
##  Mean   : 8536   Mean   : 6.027   Mean   : 713.8    Mean   :419.5
##  3rd Qu.:11388   3rd Qu.: 8.020   3rd Qu.: 783.0    3rd Qu.:490.0
##  Max.   :22770   Max.   :17.540   Max.   :1265.0    Max.   :796.0
##  total_mins_in_bed
##  Min.   : 61.0
##  1st Qu.:406.0
##  Median :463.0
##  Mean   :458.8
##  3rd Qu.:526.0
##  Max.   :961.0
```

Taking a look at the summary statistics, some notable points are:

- The average amount of steps per day for these Fitbit users is 8536, while the average for most Americans is 3000-4000 according to Mayo Clinic (https://www.mayoclinic.org/healthy-lifestyle/fitness/in-depth/10000-steps/art-20317391#:~:text=The%%2020average%20American%20walks%203%2C000,a%20day%20every%20two%20weeks).
- There appears to be someone who had 1265 sedentary minutes, meaning they were considered inactive for 21 hours. It is likely that the Fitbit did not recognize that the user was sleeping.
- On average, users slept for 7 hours per night, and spent an additional 39.3 minutes in bed.

```
ggplot(data=df) + geom_point(mapping = aes(x=total_steps, y=calories))
```




```
cor(x=df$total_steps, y=df$calories)
```

```
## [1] 0.5915681
```

There is a medium positive correlation between total steps and calories burned. This is an interesting observation that could be used in a marketing campaign about getting steps up and wearing a Bellabeat to track them.

```
df %>%
  nrow()
```

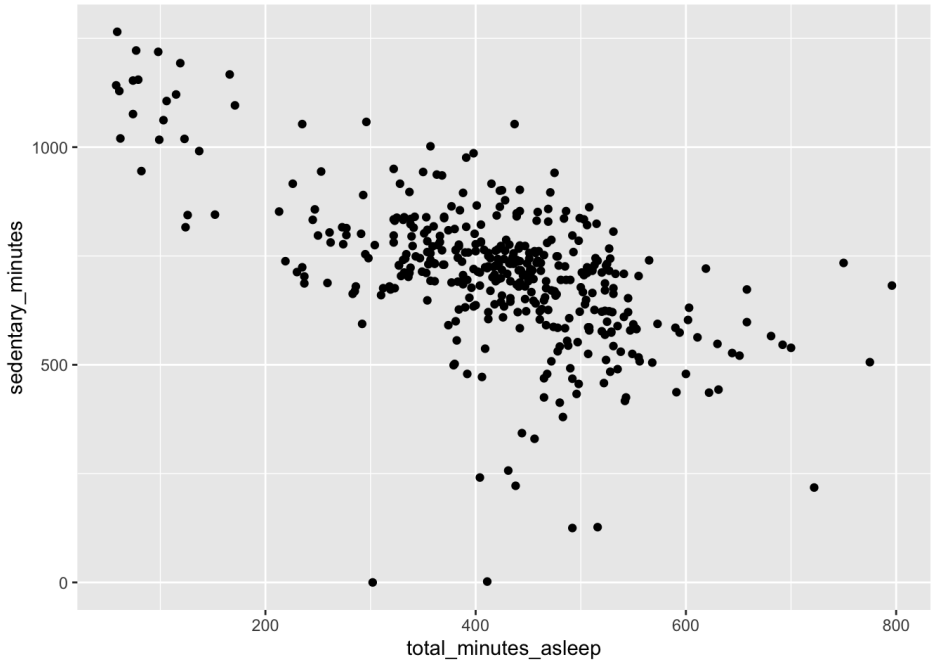
```
## [1] 940
```

```
df %>%
  filter(-total_distance==tracker_distance) %>%
  nrow()
```

```
## [1] 78
```

The tracker distance column shows how much of the user's activity (specifically distance) was tracked by the Fitbit device. Only 78 rows have different data in the distance and tracker distance column, meaning 8.3% of enties have utilized additional log uploading features.

```
df %>%
  filter(total_minutes_asleep != 0) %>%
  ggplot() + geom_point(mapping = aes(x=total_minutes_asleep, y=sedentary_minutes))
```



Interestingly, it looks like there is a negative correlation between how long users sleep and how many minutes they spend inactive throughout the day. Meaning, the longer they sleep, the less time they are inactive. There is a cluster in the top left of the graph that has very little sleep and high sedentary minutes. Assuming the Fitbit automatically detects sleep, there seems to be many instances of this detection failing.

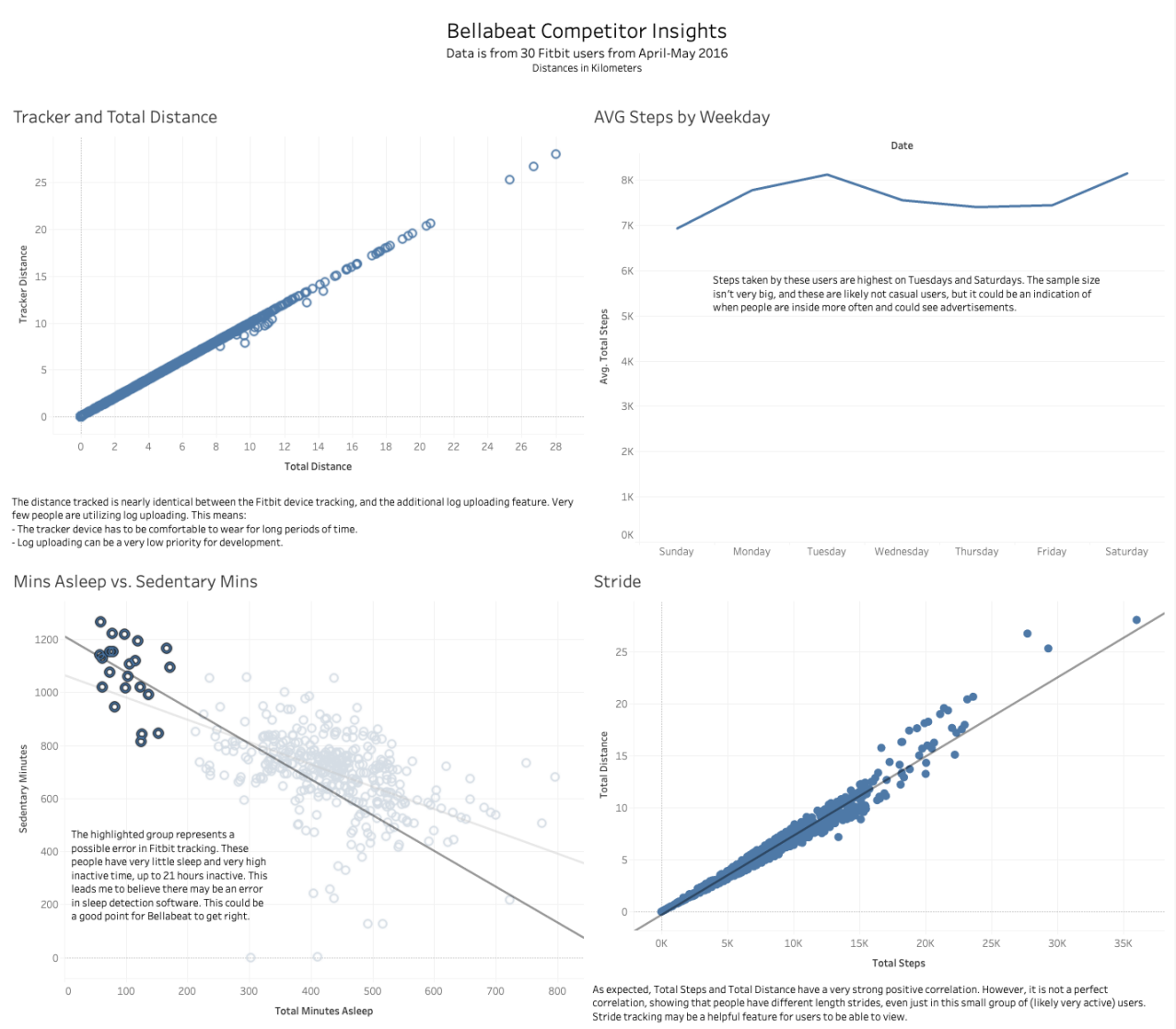
Share

Exporting newly transformed dataframe to .csv for use in Tableau.

```
write.csv(df, "merged_fitbit_data.csv")
```

Here is a preview of the Tableau dashboard, which will be included with this file. If the image is difficult to read in the report, I've included the PNG in the files.

```
knitr::include_graphics('BellabeatDashboard.png', dpi = 96)
```



Act

To sum up the points from the visual:

- Few people are using log uploading, so little resources should be allocated to developing/maintaining that feature compared to others.
    - This also hints that people are wearing their device for most of the day, so comfort has to be a top priority.
- Users are most active on Tuesdays and Saturdays, meaning they may see/hear more advertisements when they are less active.
- A possible downfall of Fitbit is the automatic sleep recognition. Bellabeat should focus on perfecting the algorithms to determine whether a user is asleep or not.
- Different users have different stride lengths, which would be an interesting metric to track on Bellabeat products.